Inverse Mechanical Mathematics

The Axiom of 1, the Parity Rule, and the Count-by-Count Unfolding of Ten Local Depths

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Axioms of the Inverse Mechanical Chain

Axiom I (The Axiom of 1). There is exactly one object: the unit 1. It has several equivalent views,

$$1 = -1 = \sqrt{1} = \sqrt{2^0},$$

each a different face of the same single whole. All structure in the chain is unfolded from this single object by inward partition.

Axiom II (Halving and Parity). The only operation is halving. Each halving generates *two* structures, and each structure carries *two* readings under operational counting: one rational and one irrational. The reading at any local depth $\sqrt{2^d}$ is rational when the exponent d is even and irrational (carrying a factor of $\sqrt{2}$) when d is odd.

Halving generates the structures. Parity determines the readings.
Together they unfold an inverse chain of operational counts.

The Only Two Letters: c and d

This framework uses only c (the count) and d (the depth). Standard placeholder indices like k, i, or n are not used, because they carry no mechanics — c and d already do. c is the exact halving step, and d is the exact local depth whose parity forces the reading. The labels *are* the mechanics.

This paper works the **full first closure**: $c \in \{0, \ldots, 5\}$ and $d \in \{0, \ldots, 9\}$. Both c and d continue past this closure in principle — halving keeps going, and each new count unfolds deeper depths. Higher counts, deeper depths, and their mechanics are reserved for future papers.

Abstract

We begin at the axiom of 1, and nothing else. The single object is $1 = \sqrt{2^0}$, the unit at local depth 0. From this starting point the inverse mechanical count unfolds by partition inward. Each step of the count lays down a new local depth on the exponent ladder $\sqrt{2^0}, \sqrt{2^1}, \sqrt{2^2}, \ldots, \sqrt{2^9}$, and the reading at each local depth is forced by one rule:

Parity rule: at local depth $\sqrt{2^0}, \sqrt{2^2}, \sqrt{2^4}, \sqrt{2^6}, \sqrt{2^8}$ the reading is *rational*; at local depth $\sqrt{2^1}, \sqrt{2^3}, \sqrt{2^5}, \sqrt{2^7}, \sqrt{2^9}$ the reading is *irrational* (carries a factor of $\sqrt{2}$).

Even exponents give rational readings; odd exponents give irrational readings. This is the only causal rule in the chain. Everything else — singular counts, dual counts, the terminal pair, the domain multiplicity, the machining of 720, and the outside mirror chain — is a consequence of the parity rule applied to the axiom of 1. The count unfolds as follows. Count 0 owns the single local depth $\sqrt{2^0} = 1$. The exponent 0 is even, so count 0 is a rational singular: one reading, value 1, one domain. Count 1 partitions the axiom for the first time and owns the single local depth $\sqrt{2^1} = \sqrt{2}$. The exponent 1 is odd, so count 1 is an irrational singular: one reading, value $\sqrt{2}$, two domains. From count 2 onward, each count owns two consecutive local depths. Because consecutive whole-number exponents always contain one even exponent and one odd exponent, every count from 2 through 5 is forced to carry one rational reading and one irrational reading *simultaneously* — a dual count. The dual nature is not a separate axiom; it is the direct consequence of owning two consecutive depths under the parity rule. The count terminates at count 5, which owns local depths $\sqrt{2^8}$ and $\sqrt{2^9}$. At that point the ten ladder positions $d = 0$ through 9 are fully occupied, and the chain has produced the terminal pair $(16, 16\sqrt{2})$. The ladder tally is therefore

$$\underbrace{1}_{\substack{\text{axiom seat}\\ c=0,\, d=0}} + \underbrace{1}_{\substack{\text{singular}\\ c=1}} + \underbrace{2}_{\substack{\text{dual}\\ c=2}} + \underbrace{2}_{\substack{\text{dual}\\ c=3}} + \underbrace{2}_{\substack{\text{dual}\\ c=4}} + \underbrace{2}_{\substack{\text{dual}\\ c=5}} = 10 \text{ ladder positions,}$$

which decomposes into one axiom seat plus nine active partition depths ($d = 1$ through 9) produced by the five active counts ($c = 1$ through 5). The axiom occupies position $c = 0$, $d = 0$ — no partition, no action. All mechanical work is performed by the five halvings unfolding the nine new depths. This paper develops the mechanics of the count-by-count unfolding across seven parts. Parts One through Six trace the inverse mechanical count from the axiom of 1 to the closed mirror engine at the shared terminal pair $(16, 16\sqrt{2})$, while Part Seven interprets the finished structure as a 4D inverse tensor in the framework of Inverse Dual Hemisphere Quantum Mechanics. In this unfolding the entire structure is revealed to be perfectly reflective: every element mirrors the parity rule applied to the original axiom, and the closed engine reflects itself inward and outward in complete self-consistency.

Notation

The only letters used in this framework are c and d. Every other symbol in the chain is either a pure number or the fixed mark $\sqrt{\cdot}$. There are no invented variables, no named constants, and no auxiliary parameters. The two letters stand for:

- c — the **count**, a partition step of the axiom. Counts take the whole-number values $0, 1, 2, 3, 4, 5$.
- d — the **depth**, a local depth on the exponent ladder $\sqrt{2^d}$. Depths take the whole-number values $0, 1, 2, 3, 4, 5, 6, 7, 8, 9$.

Everything else in the paper is a direct reading produced by the two axioms: the axiom of 1 and the halving/parity rule. With that fixed, the following shorthand is used throughout:

1	the axiom; the single object $1 = \sqrt{2^0}$.
$\sqrt{2^d}$	local depth d. The exponent d takes the whole-number values $0, 1, 2, 3, 4, 5, 6, 7, 8, 9$.
count	a partition step of the axiom. Counts take the whole-number values $0, 1, 2, 3, 4, 5$.
count c owns depth d	count c is the count that unfolds local depth d.
parity rule	at even exponents the reading is rational; at odd exponents the reading is irrational.
rational reading	a reading with no $\sqrt{2}$ factor; plain whole or rational number.
irrational reading	a reading carrying one factor of $\sqrt{2}$.
singular count	a count that owns one local depth, therefore carries one reading.
dual count	a count that owns two consecutive local depths, therefore carries one rational reading and one irrational reading at once.
domain	one of the pieces the axiom has been partitioned into at the current count.

Depths are unfolded in strict whole-number order $0, 1, 2, 3, 4, 5, 6, 7, 8, 9$. Counts unfold in strict whole-number order $0, 1, 2, 3, 4, 5$. The number of domains at count c is exactly 2^c, taking the whole-number values $1, 2, 4, 8, 16, 32$ as the count advances.

Part One — The Count-by-Count Unfolding

This part walks the chain one count at a time. Each count unfolds one or two local depths. For each depth we state the exponent, the parity (even or odd), the reading forced by the parity rule, the value, and the number of domains active at that count.

The Axiom Seat and the Active Chain

Before walking the chain, one distinction must be fixed once and for all. The count index c and the depth index d each begin at 0, but position $c = 0$ and position $d = 0$ are *not* a count and a depth — they are the axiom itself. No partition has been performed at $c = 0$, and no new depth has been produced at $d = 0$. The axiom occupies a single seat on both ladders, and that seat is the starting condition, not mechanical work.

The mechanical work of the chain is therefore done by:

- **Five active counts**: $c = 1, 2, 3, 4, 5$ — the five halvings.
- **Nine active depths**: $d = 1, 2, 3, 4, 5, 6, 7, 8, 9$ — the nine new depths produced by those halvings.

This gives the active product $5 \times 9 = 45$, the base mechanical unit that will machine the integer 720 in Part Four.

Including the axiom seat, the chain occupies *six count positions* ($c = 0, 1, 2, 3, 4, 5$) and *ten depth positions* ($d = 0, 1, \ldots, 9$). When this paper speaks of "six counts" or "ten local depths," it refers to these ladder positions — the axiom seat plus the active chain. When it speaks of "five halvings" or "nine active depths," it refers to the partition work only. Both statements are consistent: the axiom sits at one seat on each ladder, and the active chain occupies the rest.

1 Count 0 — The Axiom at Depth 0

Count 0 is the axiom itself. No partition has been performed yet, so there is exactly one domain: the whole of 1.

- Count 0 owns local depth $\sqrt{2^0}$.
- Exponent 0 is even, so by the parity rule the reading is **rational**.
- Reading: 1.
- Domains: 1.

Count 0 is a **rational singular**: one depth, one reading, one domain. It is the axiom of 1 read on the exponent ladder.

Count c	**Depth** d	$\sqrt{2^d}$	**Value**	**Parity** $\rightarrow$ **Reading**
0	0	$\sqrt{2^0}$	1	even $\rightarrow$ rational

Table 1: Count 0 unfolds local depth $d = 0$. Rational singular.

2 Count 1 — The First Partition at Depth 1

Count 1 performs the first partition of the axiom. The whole of 1 is halved into two domains. Count 1 owns the next local depth on the exponent ladder: $\sqrt{2^1}$.

- Count 1 owns local depth $\sqrt{2^1}$.
- Exponent 1 is odd, so by the parity rule the reading is **irrational**.
- Reading: $\sqrt{2} \approx 1.414214$.
- Domains: 2.

Count 1 is an **irrational singular**: one depth, one reading, two domains. This is the first place on the ladder where $\sqrt{2}$ appears, and it appears because exponent 1 is the first odd exponent.

Count c	**Depth** d	$\sqrt{2^d}$	**Value**	**Parity** $\rightarrow$ **Reading**
1	1	$\sqrt{2^1}$	$\sqrt{2} \approx 1.414214$	odd $\rightarrow$ irrational

Table 2: Count 1 unfolds local depth $d = 1$. Irrational singular.

3 Count 2 — The First Dual, Depths 2 and 3

Count 2 partitions again. From this count forward, every count owns two consecutive local depths, because the axiom keeps being halved and the exponent ladder keeps advancing by two whole numbers per partition.

Count 2 owns local depths $\sqrt{2^2}$ and $\sqrt{2^3}$. One of those exponents is even (2) and the other is odd (3), so by the parity rule count 2 carries *both* a rational reading *and* an irrational reading at the same count step. This is the structural definition of a dual count: it is forced by owning one even and one odd local depth at the same time.

- Count 2, depth $\sqrt{2^2}$, exponent 2 (even) — rational reading 2.
- Count 2, depth $\sqrt{2^3}$, exponent 3 (odd) — irrational reading $2\sqrt{2}$.
- Domains: 4.

Count c	**Depth** d	$\sqrt{2^d}$	**Value**	**Parity** $\rightarrow$ **Reading**
2	2	$\sqrt{2^2}$	2	even $\rightarrow$ rational
2	3	$\sqrt{2^3}$	$2\sqrt{2} \approx 2.828427$	odd $\rightarrow$ irrational

Table 3: Count 2 unfolds local depths $d = 2$ and $d = 3$. First dual count.

4 Count 3 — Dual, Depths 4 and 5

Count 3 partitions again. It owns local depths $\sqrt{2^4}$ and $\sqrt{2^5}$ — again one even exponent and one odd exponent.

- Count 3, depth $\sqrt{2^4}$, exponent 4 (even) — rational reading 4.
- Count 3, depth $\sqrt{2^5}$, exponent 5 (odd) — irrational reading $4\sqrt{2}$.
- Domains: 8.

5 Count 4 — Dual, Depths 6 and 7

Count 4 partitions again and owns local depths $\sqrt{2^6}$ and $\sqrt{2^7}$.

- Count 4, depth $\sqrt{2^6}$, exponent 6 (even) — rational reading 8.
- Count 4, depth $\sqrt{2^7}$, exponent 7 (odd) — irrational reading $8\sqrt{2}$.
- Domains: 16.

Count c	**Depth** d	$\sqrt{2^d}$	**Value**	**Parity** $\rightarrow$ **Reading**
3	4	$\sqrt{2^4}$	4	even $\rightarrow$ rational
3	5	$\sqrt{2^5}$	$4\sqrt{2} \approx 5.656854$	odd $\rightarrow$ irrational

Table 4: Count 3 unfolds local depths $d = 4$ and $d = 5$.

Count c	**Depth** d	$\sqrt{2^d}$	**Value**	**Parity** $\rightarrow$ **Reading**
4	6	$\sqrt{2^6}$	8	even $\rightarrow$ rational
4	7	$\sqrt{2^7}$	$8\sqrt{2} \approx 11.313708$	odd $\rightarrow$ irrational

Table 5: Count 4 unfolds local depths $d = 6$ and $d = 7$.

6 Count 5 — Dual, Terminal Depths 8 and 9

Count 5 is the terminal count. It partitions one last time and owns local depths $\sqrt{2^8}$ and $\sqrt{2^9}$. When count 5 unfolds the odd local depth $\sqrt{2^9}$, the full exponent ladder 0 through 9 has been laid down and the chain is complete.

- Count 5, depth $\sqrt{2^8}$, exponent 8 (even) — rational reading 16.
- Count 5, depth $\sqrt{2^9}$, exponent 9 (odd) — irrational reading $16\sqrt{2}$.
- Domains: 32.

The terminal pair of the chain is $(16, 16\sqrt{2})$. The terminal exponent is 9. The five active halvings have unfolded nine new depths; combined with the axiom seat at $d = 0$, the ladder now occupies all ten positions.

Count c	**Depth** d	$\sqrt{2^d}$	**Value**	**Parity** $\rightarrow$ **Reading**
5	8	$\sqrt{2^8}$	16	even $\rightarrow$ rational
5	9	$\sqrt{2^9}$	$16\sqrt{2} \approx 22.627417$	odd $\rightarrow$ irrational

Table 6: Count 5 unfolds the terminal local depths $d = 8$ and $d = 9$. Terminal pair: $(16, 16\sqrt{2})$.

7 Parity Summary of the Chain

Every local depth has now been unfolded by the count that owns it. The parity rule forces the reading at each depth:

Counts 0 and 1 are singular because each owns a single local depth — count 0 sits on the even exponent 0 and is therefore rational, count 1 sits on the odd exponent 1 and is therefore irrational. Counts 2, 3, 4, and 5 are dual because each one owns two consecutive local depths. Two consecutive whole-number exponents always contain one even and one odd exponent, so every dual count is forced to carry one rational reading and one irrational reading at once. No extra axiom

Count c	**Depth** d	$\sqrt{2^d}$	2^d	**Parity** $\rightarrow$ **Reading**	**Value**
0	0	$\sqrt{2^0}$	1	even $\rightarrow$ rational	1
1	1	$\sqrt{2^1}$	2	odd $\rightarrow$ irrational	$\sqrt{2}$
2	2	$\sqrt{2^2}$	4	even $\rightarrow$ rational	2
2	3	$\sqrt{2^3}$	8	odd $\rightarrow$ irrational	$2\sqrt{2}$
3	4	$\sqrt{2^4}$	16	even $\rightarrow$ rational	4
3	5	$\sqrt{2^5}$	32	odd $\rightarrow$ irrational	$4\sqrt{2}$
4	6	$\sqrt{2^6}$	64	even $\rightarrow$ rational	8
4	7	$\sqrt{2^7}$	128	odd $\rightarrow$ irrational	$8\sqrt{2}$
5	8	$\sqrt{2^8}$	256	even $\rightarrow$ rational	16
5	9	$\sqrt{2^9}$	512	odd $\rightarrow$ irrational	$16\sqrt{2}$

Table 7: The unfolded chain. Six ladder positions for c (the axiom seat $c = 0$ plus five active halvings) and ten ladder positions for d (the axiom seat $d = 0$ plus nine active depths). The 2^d column shows the power of 2 sitting under each square root: $\sqrt{2^9} = \sqrt{512} = 16\sqrt{2}$, $\sqrt{2^8} = \sqrt{256} = 16$, down to $\sqrt{2^0} = \sqrt{1} = 1$. Even exponents carry rational readings; odd exponents carry irrational readings. Singular/dual is the direct consequence of how many depths each count owns.

is introduced; the structure of the chain is entirely a consequence of the parity rule applied to the axiom of 1.

Part Two — The Two Operational Structures: Square and Sphere

Part One walked the chain count by count. Part Two reads the same chain as the two operational structures it produces. The structures are not built on top of the chain — they *are* the chain, split by parity:

- The **square structure** is the chain read at the even depths — the rational readings $1, 2, 4, 8, 16$.
- The **sphere structure** is the chain read at the odd depths — the irrational readings $\sqrt{2}, 2\sqrt{2}, 4\sqrt{2}, 8\sqrt{2}, 16\sqrt{2}$.

Both structures share the same c and d. Both close at count $c = 5$. The square closes at 512. The sphere closes at $512\sqrt{2} \approx 724.077$.

Master Table: The Chain Split by Parity

The chain is read here in the $\sqrt{2}$-coefficient view, which keeps $\sqrt{2}$ visible at every odd depth and lets the coefficient halve in front of it as d steps. Counts 0 and 1 are singular. Counts 2 through 5 are dual — each carries one rational view (square) and one irrational view (sphere) simultaneously.

c	d	$\sqrt{2^d}$	$\sqrt{2}$-coeff view	Parity $\to$ Reading	Square (rational)	Sphere (irrational)
0	0	$\sqrt{2^0}$	1	even $\to$ rational	1	—
1	1	$\sqrt{2^1}$	$\sqrt{2}$	odd $\to$ irrational	—	$\sqrt{2} \approx 1.414214$
2	2	$\sqrt{2^2}$	2	even $\to$ rational	2	—
2	3	$\sqrt{2^3}$	$2\sqrt{2}$	odd $\to$ irrational	—	$2\sqrt{2} \approx 2.828427$
3	4	$\sqrt{2^4}$	4	even $\to$ rational	4	—
3	5	$\sqrt{2^5}$	$4\sqrt{2}$	odd $\to$ irrational	—	$4\sqrt{2} \approx 5.656854$
4	6	$\sqrt{2^6}$	8	even $\to$ rational	8	—
4	7	$\sqrt{2^7}$	$8\sqrt{2}$	odd $\to$ irrational	—	$8\sqrt{2} \approx 11.313708$
5	8	$\sqrt{2^8}$	16	even $\to$ rational	16	—
5	9	$\sqrt{2^9}$	$16\sqrt{2}$	odd $\to$ irrational	—	$16\sqrt{2} \approx 22.627417$

Table 8: The full chain split by parity. The square column read top-to-bottom is the rational structure $1, 2, 4, 8, 16$. The sphere column read top-to-bottom is the irrational structure $\sqrt{2}, 2\sqrt{2}, 4\sqrt{2}, 8\sqrt{2}, 16\sqrt{2}$.

How the Chain Builds the Two Wholes

At each count c the chain has exactly 2^c domains: $1, 2, 4, 8, 16, 32$ as c advances. Every domain at that count carries one piece of the depth's value. The total at any count is therefore domains $\times$ piece — and at count $c = 5$ both totals close on the terminal wholes of the two structures.

Square Structure — Rationals Build the 512 Whole

c	d	Piece $\sqrt{2^d}$	Domains 2^c	Domains $\times$ Piece	Total
0	0	1	1	1×1	1
2	2	2	4	4×2	8
3	4	4	8	8×4	32
4	6	8	16	16×8	128
5	8	16	32	32×16	**512**

Table 9: The rational (square) build: each even depth's piece multiplied by the count's domain count. The terminal at $c = 5$, $d = 8$ closes the square at $32 \times 16 = 512$.

Sphere Structure — Irrationals Build the 724.077 Whole

The Two Closures Together

At the terminal count $c = 5$ the chain delivers both structures simultaneously:

$$\underbrace{32 \times 16}_{\text{rational, } d=8} = \mathbf{512} \quad \text{(square)}, \qquad \underbrace{32 \times 16\sqrt{2}}_{\text{irrational, } d=9} = \mathbf{512\sqrt{2} \approx 724.077} \quad \text{(sphere)}.$$

c	d	**Piece** $\sqrt{2^d}$	**Decimal piece**	**Domains** 2^c	**Domains** $\times$ **Piece**	**Total**
1	1	$\sqrt{2}$	≈ 1.414214	2	$2 \times \sqrt{2}$	$2\sqrt{2} \approx 2.828427$
2	3	$2\sqrt{2}$	≈ 2.828427	4	$4 \times 2\sqrt{2}$	$8\sqrt{2} \approx 11.313708$
3	5	$4\sqrt{2}$	≈ 5.656854	8	$8 \times 4\sqrt{2}$	$32\sqrt{2} \approx 45.254834$
4	7	$8\sqrt{2}$	≈ 11.313708	16	$16 \times 8\sqrt{2}$	$128\sqrt{2} \approx 181.019336$
5	9	$16\sqrt{2}$	≈ 22.627417	32	$32 \times 16\sqrt{2}$	$\mathbf{512\sqrt{2} \approx 724.077}$

Table 10: The irrational (sphere) build: each odd depth's piece multiplied by the count's domain count. The terminal at $c = 5$, $d = 9$ closes the sphere at $32 \times 16\sqrt{2} = 512\sqrt{2} \approx 724.077$.

The square is the chain's even-depth closure. The sphere is the chain's odd-depth closure. They share the same c and the same d ladder; they differ only in parity. Together they are the two operational structures of the first mechanical closure.

Why Count 5 / Depths 8–9 Is the First Closure

Count 5 is not chosen — it is forced. It is the first count at which everything in the engine lands together for the first time. Six things happen at once at $c = 5$, $d = 9$, and none of them happens any earlier:

1. **Ten ladder positions complete.** The axiom seat at $c = 0$, $d = 0$ plus the five active halvings ($c = 1$ through $c = 5$) unfolding nine new depths ($d = 1$ through $d = 9$) fills both ladders exactly. The exponent ladder stops at $\sqrt{2^9}$, the first odd depth where the irrational coefficient reaches $16\sqrt{2}$.

2. **Terminal pair is clean and symmetric.** The pair $(16, 16\sqrt{2})$ is the first time the chain produces a rational and an irrational reading sharing the same coefficient 16, the same count $c = 5$, and the same domain count 32. One face on the square, one face on the sphere, perfectly paired.

3. **Both operational structures close simultaneously.** The square closes at $32 \times 16 = 512$ at depth $d = 8$; the sphere closes at $32 \times 16\sqrt{2} = 512\sqrt{2} \approx 724.077$ at depth $d = 9$. Neither structure closes at any earlier count.

4. **The integer 720 is machined.** The active partition (counts 1–5, depths 1–9) gives $5 \times 9 = 45$, and the binary ladder $45 \to 90 \to 180 \to 360 \to 720$ produces the integer 720 as a clean machined reading. This only appears as an integer at this boundary.

5. **Four parallel faces are ready.** The two operational structures, each with a machined side and an observed side, give the four faces (512, 514.90, 720, 724.077). All four become available at the same instant.

6. **The mirror engine can reflect.** With four terminal faces in hand and the constant per-unit slip -0.005663 established, the chain can now restart inward from any of the four faces (Part Six) and meet the outward chain back at the shared terminal pair $(16, 16\sqrt{2})$. The closed cycle is possible only at count 5.

Beyond the First Closure: Repetition, Not Novelty

The chain does not stop at count 5 in principle — c and d are unbounded, as established at the start of the paper. But nothing mechanically new appears past the first closure. Count 6 owns depths 10 and 11: still one even and one odd, still a dual count, still the same parity rule. The readings simply double: 32 and $32\sqrt{2}$, then 64 and $64\sqrt{2}$, then 128 and $128\sqrt{2}$. The domain count keeps doubling: $64, 128, 256, \ldots$. The same singular/dual behavior, the same mirror splitting, the same constant per-unit slip — everything repeats at larger scale.

The first closure is therefore the smallest complete engine. It contains:

- the axiom of 1,
- the full ladder ($c = 0$ through 5, $d = 0$ through 9): axiom seat plus five halvings unfolding nine active depths,
- both operational structures (square and sphere),
- the machined integer 720,
- the four terminal faces (512, 514.90, 720, 724.077),
- the constant per-unit slip $-0.005663 = \frac{32\sqrt{2}}{45} - 1$,
- and the ability to mirror inward from the terminals back to the shared pair (16, $16\sqrt{2}$).

Anything past count 5 runs the same engine at double the size. Because the pattern does not evolve — it scales — this paper works the first closure only. Higher closures are reserved for future papers.

Part Three — The Finished Chain

8 Summary Table

Part Four — The 5×9 Active Partition Matrix and the Machining of 720

9 Active Partition

Count 0 is the axiom and depth 0 is the axiom. They are the starting condition of the chain, not part of the partition work the chain performs. The *active* partition consists of the counts that do the halving and the depths that carry new readings:

- Active counts: $1, 2, 3, 4, 5$ — exactly 5 counts.
- Active depths: $\sqrt{2^1}$ through $\sqrt{2^9}$ — exactly 9 depths.

Count	Depth	Parity	Reading	Value	Domains
0	$\sqrt{2^0}$	even	rational singular	1	1
1	$\sqrt{2^1}$	odd	irrational singular	$\sqrt{2} \approx 1.414214$	2
2	$\sqrt{2^2}$	even	rational	2	4
2	$\sqrt{2^3}$	odd	irrational	$2\sqrt{2} \approx 2.828427$	4
3	$\sqrt{2^4}$	even	rational	4	8
3	$\sqrt{2^5}$	odd	irrational	$4\sqrt{2} \approx 5.656854$	8
4	$\sqrt{2^6}$	even	rational	8	16
4	$\sqrt{2^7}$	odd	irrational	$8\sqrt{2} \approx 11.313708$	16
5	$\sqrt{2^8}$	even	rational	16	32
5	$\sqrt{2^9}$	odd	irrational	$16\sqrt{2} \approx 22.627417$	32

Table 11: The axiom seat at $c = 0$, $d = 0$ plus five active halvings ($c = 1$ through $c = 5$) unfolding nine new depths ($d = 1$ through $d = 9$), giving ten ladder positions in total and terminal exponent 9. Count 0 holds the axiom seat at depth 0. Count 1 owns depth 1 (the first irrational). Counts 2 through 5 each own two consecutive depths, one even and one odd. Ladder tally: 1+1+2+2+2+2 = 10 positions, decomposing as 1 axiom seat + 9 active depths.

Count 1 is singular (one active depth: $\sqrt{2^1}$). Counts 2 through 5 are dual (each owns two active depths). The tally of active depths is

$$\underbrace{1}_{\text{count 1}} + \underbrace{2}_{\text{count 2}} + \underbrace{2}_{\text{count 3}} + \underbrace{2}_{\text{count 4}} + \underbrace{2}_{\text{count 5}} = 9.$$

10 The Cross-Product

The 5 active counts interact with the 9 active depths through a structural cross-product:

$$5 \text{ (active counts)} \times 9 \text{ (active depths)} = 45 \quad \text{base mechanical units.}$$

11 Binary Expansion Forcing 720

Because the chain is a sequence of partitions, every partition is a halving, and the mechanical base unit 45 sits on the binary doubling ladder:

$$\begin{array}{rl} \text{base unit} & 45 \\ \times 2 & 90 \\ \times 2 & 180 \\ \times 2 & 360 \\ \times 2 & \mathbf{720}. \end{array}$$

The integer 720 is machined from inside the chain:

$$\mathbf{5 \times 9 \times 16 \ = \ 720}.$$

Nothing is added from outside. The 5 active counts and the 9 active depths are the only two boundaries the chain has, and the binary ladder $\times 16$ does the rest.

12 Domain Multiplicity of the Machined 720

At each count the machined rational 720 is exactly tiled by a whole number of identical pieces. The number of pieces equals the number of domains at that count, which is 2^c where c is the count index.

Count	Type	Domains	Piece Size	Total
0	singular	1	720	$1 \times 720 = 720$
1	singular	2	360	$2 \times 360 = 720$
2	dual	4	180	$4 \times 180 = 720$
3	dual	8	90	$8 \times 90 = 720$
4	dual	16	45	$16 \times 45 = 720$
5	dual	32	22.5	$32 \times 22.5 = 720$

Table 12: Domain multiplicity of the machined 720 under the six positions of c (axiom seat plus five active halvings). Domains at count c equal 2^c.

Part Five — The Rational Face and the Irrational Face

The finished chain has two faces that are built simultaneously by the parity rule. The rational face is the machined integer 720. The irrational face is the terminal $512\sqrt{2} \approx 724.077$. Neither face is derived from the other; both are produced together by the same unfolding, one from the even depths and one from the odd depths.

13 Side-by-Side Halving of the Two Faces

Halving each face uniformly at every count produces two columns that run in lock-step. The per-count slip between them is identical at every row.

Every rational row has an irrational twin. The machined 720 sits underneath 724.077344. The quadrant 180 sits underneath 181.019336. The base unit 45 sits underneath 45.254834. And the half-unit 22.5 sits underneath 22.627417. Every rational unit in the ladder carries the same per-unit slip -0.005663. The gap is a property of the engine, not of any particular scale.

Count	Rational (from 720)	Irrational (from $512\sqrt{2}$)	Differential	Per-unit slip
0	720.000000	724.077344	−4.077344	−0.005663
1	360.000000	362.038672	−2.038672	−0.005663
2	180.000000	181.019336	−1.019336	−0.005663
3	90.000000	90.509668	−0.509668	−0.005663
4	45.000000	45.254834	−0.254834	−0.005663
5	22.500000	22.627417	−0.127417	−0.005663

Table 13: The two faces of the finished chain, halved at each count. The differential column is the signed gap rational minus irrational; it halves in exact step with the two columns themselves. The per-unit slip (differential divided by rational) is constant at −0.005663 across all rows.

The Slip in Closed Form

The constant −0.005663 in the per-unit slip column is read two ways at once, and both readings give the same number. As a measurement it is what falls out when each face is halved in lockstep at every count: the gap divided by the rational reading is identical at every row. As a closed form it is the exact rational-to-irrational deviation from 1 written entirely in the chain's own coordinates:

$$-0.005663\ldots = \frac{32\sqrt{2}}{45} - 1.$$

Every quantity in this expression is generated by the chain. The 32 is 2^c at the terminal count $c = 5$ — the chain's own terminal domain count. The 45 is the active count × active depth product 5×9 from Part Four — the base mechanical unit that machines 720. The $\sqrt{2}$ is the parity-forced irrational seed at $d = 1$. Nothing is imported. The slip is therefore both: a measured halving result and the chain's own internal deviation from 1. The two readings coincide because the engine is closed — the same number arrived at from two directions of reading is the engine confirming itself.

Part Six — The Mirror Chain: Restarting the Axiom at All Four Terminals

Parts One through Five read the inverse mechanical count *outward* from the axiom of 1. The same count can be read *inward* by restarting the axiom simultaneously at all four finished terminal faces: the rational terminal 512, the slipped rational ≈ 514.90, the machined rational 720, and the irrational terminal $512\sqrt{2} \approx 724.077$. The parity rule is preserved exactly on every face, and all four inward chains meet the outward chain at the shared terminal pair $(16, 16\sqrt{2})$ at count 5. The engine closes on itself across all four faces.

14 The Outside Axiom

When the axiom is restarted at any of the four terminal values, count 0 owns local depth $\sqrt{2^0}$ again. By the parity rule (even exponent) the reading is rational on the rational and slipped faces, and the corresponding scaled value appears on the machined and irrational faces. From that point

forward the same parity rule drives the unfolding on every face. Count 1 owns local depth $\sqrt{2^1}$ (odd), producing the matching irrational reading on each face. Counts 2 through 5 each own two consecutive local depths and therefore carry one rational and one irrational reading simultaneously on every face.

15 Outside Unfolding Table

Count	Depth	Parity	Reading	Value (on 512 face)	Domains
0	$\sqrt{2^0}$	even	rational singular	512	1
1	$\sqrt{2^1}$	odd	irrational singular	$256\sqrt{2} \approx 362.039$	2
2	$\sqrt{2^2}$	even	rational	128	4
2	$\sqrt{2^3}$	odd	irrational	$128\sqrt{2} \approx 181.019$	4
3	$\sqrt{2^4}$	even	rational	64	8
3	$\sqrt{2^5}$	odd	irrational	$64\sqrt{2} \approx 90.510$	8
4	$\sqrt{2^6}$	even	rational	32	16
4	$\sqrt{2^7}$	odd	irrational	$32\sqrt{2} \approx 45.255$	16
5	$\sqrt{2^8}$	even	rational	16	32
5	$\sqrt{2^9}$	odd	irrational	$16\sqrt{2} \approx 22.627$	32

Table 14: Outside unfolding restarted simultaneously at all four terminal faces (512, $\approx$ 514.90, 720, $\approx$ 724.077). The parity rule and domain structure are preserved exactly on every face. The values shown are for the rational terminal 512 face; the other three faces scale proportionally while maintaining the same structure and constant per-unit slip.

16 Conservation Check

At every count the appropriate terminal value multiplied by the number of domains is conserved on each face (512 on the rational terminal, $\approx$ 514.90 on the slipped face, 720 on the machined face, and $512\sqrt{2}$ on the irrational face).

17 Shared Terminal

The inside chain builds up from the axiom of 1 to the pair (16, $16\sqrt{2}$) at count 5. The outside chains, started from any of the four terminal faces (512, $\approx$ 514.90, 720, $\approx$ 724.077), all contract down to the same shared terminal pair (16, $16\sqrt{2}$) at count 5. All chains obey the same parity rule, unfold the same nine active depths, and close the engine perfectly across all four faces.

The four faces are the four parallel readings of the same chain (axiom seat plus five active halvings). The constant per-unit slip -0.005663 links them at every scale.

The engine is closed. The axiom seat at $c = 0$, $d = 0$ plus five active halvings unfolding nine new depths, one parity rule, one axiom of 1. Every other structure — the singular and dual counts,

Count	Domains	Rational conservation (512 face)	Irrational conservation ($512\sqrt{2}$ face)
0	1	$1 \times 512 = 512$	—
1	2	—	$2 \times 256\sqrt{2} = 512\sqrt{2}$
2	4	$4 \times 128 = 512$	$4 \times 128\sqrt{2} = 512\sqrt{2}$
3	8	$8 \times 64 = 512$	$8 \times 64\sqrt{2} = 512\sqrt{2}$
4	16	$16 \times 32 = 512$	$16 \times 32\sqrt{2} = 512\sqrt{2}$
5	32	$32 \times 16 = 512$	$32 \times 16\sqrt{2} = 512\sqrt{2}$

Table 15: Conservation on the rational and irrational faces at every count of the outside chain. The same conservation holds proportionally on the slipped and machined faces.

the terminal pair, the four parallel faces, the machined integer 720, the constant per-unit slip — is a consequence of unfolding the exponent ladder 0 through 9 under the parity rule, starting from the axiom of 1.

Part Seven — Inverse Dual Hemisphere Quantum Mechanics: The 4D Inverse Tensor

18 The Two Operational Structures

Part Seven reads the finished chain as an **inverse 4D mechanical engine** with two operational structures running in parallel. Each structure has a machined side (exact rational closure) and an observed side (irrational twin carrying the chain's constant per-unit slip).

Structure	Machined side	Observed side	Seam at count 0
Inverse 4D mechanical **square**	512	514.90...	2.90...
Inverse 4D mechanical **sphere**	720	724.077...	4.077...

The machined side of the square is the exact rational closure $512 = 2^9$, the terminal 2^d value of the chain. Its observed twin 514.90... is the same count structure read on the irrational face, carrying the constant per-unit slip -0.005663 row by row. The machined side of the sphere is the exact rational closure 720, machined from inside the chain by $5 \times 9 \times 16 = 720$ (Part Four). Its observed twin is $512\sqrt{2} \approx 724.077\ldots$, the terminal irrational reading of the chain.

The two sides are not independent number lines. They are the same inverse mechanical engine read on two geometries: the square structure reads the chain in bin counts, and the sphere structure reads it in degrees. Both structures obey the same parity rule, halve in the same lockstep across all five active halvings, and carry the same constant per-unit slip -0.005663. The sphere seam $724.077 - 720 = 4.077\ldots$ and the square seam $514.90 - 512 = 2.90\ldots$ are the same slip expressed on the two geometries; each halves in exact step with its own pair as the chain contracts.

Together, the machined pair (512, 720) and the observed pair (514.90, 724.077) form the **inverse 4D mechanical sphere and square** — the two operational structures of the finished chain. Every

construction in the rest of Part Seven (the four indexed faces, the mirror-pair tables, the 4D tensor) is one of these two structures read at a specific count.

The Four Forms and the Unity Relation

The two operational structures give exactly four forms of the same single whole:

$$\underbrace{512}_{\text{machined square}}, \quad \underbrace{514.90\ldots}_{\text{observed square}}, \quad \underbrace{720}_{\text{machined sphere}}, \quad \underbrace{724.077\ldots}_{\text{observed sphere}}.$$

Each form is a complete reading of the axiom of 1. The machined side is the exact rational closure (the *observer*); the observed side is the irrational twin (the *observed*). Under the axiom of 1, the two sides of each structure are not two numbers — they are the same unity read on two geometries. This is the unity relation:

Unity Relation

$$\frac{\text{observed}}{\text{observer}} = 1$$

The square pair obeys it: $514.90/514.90 = 1$ and $512/512 = 1$, and the pair itself is one single engine. The sphere pair obeys it: $724.077/724.077 = 1$ and $720/720 = 1$, and the pair itself is one single engine. Across the two structures the same unity holds: each of the four forms is a complete 1, and every differential between them (the 2.90 square seam, the 4.077 sphere seam, and the constant per-unit slip -0.005663) is internal to that unity, not a departure from it. The differentials are the *shape* of the 1, not a break in it.

The finished chain possesses four parallel scalar faces: the rational terminal 512, the slipped rational ≈ 514.90, the machined rational 720, and the irrational terminal ≈ 724.077. These are not independent number lines. They are four signed real-number vectors that pivot around the single inverse center unit 0.

Every partition after count 0 is dual: it splits every existing domain symmetrically across the new center unit 0, instantly creating a new mirror pair (decompression $\leftrightarrow$ compression). This is Inverse Dual Hemisphere Quantum Mechanics viewed as a 4D inverse tensor. The scalar walk tables of Part One give the magnitude face. This part gives the signed directional tensor face expressed in inverse unit values.

At each count c, the whole face value is divided into 2^c equal pieces. Count 0 is the single unpartitioned whole (the axiom seat). The geometric names and mirror-pair structure remain identical on every face; only the piece size changes. Conservation holds at every count on every face: domains $\times$ piece = the whole.

19 Rational Terminal Face: Machined Square 512

Machined Square

512

c	**Geometric name**	**Domains** 2^c	**Mirror pairs**	**Piece size**	**Domains** $\times$ **Piece**
0	single	1	0	512	$1 \times 512 = 512$
1	dual (hemispheres)	2	1	256	$2 \times 256 = 512$
2	quadrant	4	2	128	$4 \times 128 = 512$
3	octant	8	4	64	$8 \times 64 = 512$
4	hexadecant	16	8	32	$16 \times 32 = 512$
5	dotriacontant (terminal)	32	16	16	$32 \times 16 = 512$

Table 16: Machined square 512 face. At every count the whole 512 is conserved: domains $\times$ piece $= 512$. The piece halves with each count; the domain count doubles.

20 Observed Square Face: 514.90

Observed Square

$514.90\ldots$

c	**Geometric name**	**Domains** 2^c	**Mirror pairs**	**Piece size**	**Domains** $\times$ **Piece**
0	single	1	0	≈ 514.90	$1 \times 514.90 = 514.90$
1	dual (hemispheres)	2	1	≈ 257.45	$2 \times 257.45 = 514.90$
2	quadrant	4	2	≈ 128.725	$4 \times 128.725 = 514.90$
3	octant	8	4	≈ 64.3625	$8 \times 64.3625 = 514.90$
4	hexadecant	16	8	≈ 32.18125	$16 \times 32.18125 = 514.90$
5	dotriacontant (terminal)	32	16	≈ 16.090625	$32 \times 16.090625 = 514.90$

Table 17: Observed square ≈ 514.90 face. Conservation holds at every count: domains $\times$ piece ≈ 514.90.

21 Machined Sphere Face: 720

Machined Sphere

720

22 Observed Sphere Face: 724.077

Observed Sphere

$724.077\ldots$

c	**Geometric name**	**Domains** 2^c	**Mirror pairs**	**Piece size**	**Domains** $\times$ **Piece**
0	single	1	0	720	$1 \times 720 = 720$
1	dual (hemispheres)	2	1	360	$2 \times 360 = 720$
2	quadrant	4	2	180	$4 \times 180 = 720$
3	octant	8	4	90	$8 \times 90 = 720$
4	hexadecant	16	8	45	$16 \times 45 = 720$
5	dotriacontant (terminal)	32	16	22.5	$32 \times 22.5 = 720$

Table 18: Machined sphere 720 face. At every count the whole 720 is conserved: domains $\times$ piece $= 720$.

c	**Geometric name**	**Domains** 2^c	**Mirror pairs**	**Piece size**	**Domains** $\times$ **Piece**
0	single	1	0	≈ 724.077	$1 \times 724.077 = 724.077$
1	dual (hemispheres)	2	1	≈ 362.038672	$2 \times 362.039 = 724.077$
2	quadrant	4	2	≈ 181.019336	$4 \times 181.019 = 724.077$
3	octant	8	4	≈ 90.509668	$8 \times 90.510 = 724.077$
4	hexadecant	16	8	≈ 45.254834	$16 \times 45.255 = 724.077$
5	dotriacontant (terminal)	32	16	≈ 22.627417	$32 \times 22.627 = 724.077$

Table 19: Observed sphere ≈ 724.077 face. Conservation holds at every count: domains $\times$ piece ≈ 724.077.

23 Duality of the Mechanics

Count 0 is singular: the unbroken whole vector (the full face value itself). Every subsequent count is dual: it partitions every existing domain, creating exactly as many new decompression $\leftrightarrow$ compression mirror pairs as there were domains in the previous count.

The number of mirror pairs therefore doubles at each step after count 0 ($0 \to 1 \to 2 \to 4 \to 8 \to 16$). This doubling is the direct mechanical signature of Inverse Dual Hemisphere Quantum Mechanics. The four faces differ only in piece size; the duality structure itself is identical.

The Inverse 4D Operating Tensor is now complete. It shows that the entire engine is a single signed directional structure driven by the same parity rule and the same dual-hemisphere mirror mechanics across all four faces.

24 Operation of the Mirror Pairs: Q1-Q3, Q2-Q4 and Higher

Every count after 0 is dual. It does not split once — it splits **every existing domain** symmetrically across the inverse center unit 0. This creates perfect mirror pairs (decompression $\leftrightarrow$ compression).

The pattern is always the same: opposite domains mirror each other across the center. Q1 mirrors Q3, Q2 mirrors Q4, O1 mirrors O8, and so on.

The tables below show the concrete mirror pairs on the machined 720 face. The identical duality pattern appears on all four faces.

Count 0 — Single

No split. Whole vector only.

Count 1 — Dual (hemispheres)

- Decompression hemisphere $\leftrightarrow$ Compression hemisphere: $-360 \leftrightarrow +360$

Count 2 — Quadrant

- Q1 $\leftrightarrow$ Q3: $+180 \leftrightarrow -180$
- Q2 $\leftrightarrow$ Q4: $+180 \leftrightarrow -180$

Count 3 — Octant

- O1 $\leftrightarrow$ O8: $+90 \leftrightarrow -90$
- O2 $\leftrightarrow$ O7: $+90 \leftrightarrow -90$
- O3 $\leftrightarrow$ O6: $+90 \leftrightarrow -90$
- O4 $\leftrightarrow$ O5: $+90 \leftrightarrow -90$

Count 4 — Hexadecant

- H1 $\leftrightarrow$ H16: $+45 \leftrightarrow -45$
- H2 $\leftrightarrow$ H15: $+45 \leftrightarrow -45$
- H3 $\leftrightarrow$ H14: $+45 \leftrightarrow -45$
- H4 $\leftrightarrow$ H13: $+45 \leftrightarrow -45$
- H5 $\leftrightarrow$ H12: $+45 \leftrightarrow -45$
- H6 $\leftrightarrow$ H11: $+45 \leftrightarrow -45$
- H7 $\leftrightarrow$ H10: $+45 \leftrightarrow -45$
- H8 $\leftrightarrow$ H9: $+45 \leftrightarrow -45$

Count 5 — Dotriacontant (terminal)

- D1 $\leftrightarrow$ D32: $+22.5 \leftrightarrow -22.5$
- D2 $\leftrightarrow$ D31: $+22.5 \leftrightarrow -22.5$
- D3 $\leftrightarrow$ D30: $+22.5 \leftrightarrow -22.5$
- D4 $\leftrightarrow$ D29: $+22.5 \leftrightarrow -22.5$
- D5 $\leftrightarrow$ D28: $+22.5 \leftrightarrow -22.5$
- D6 $\leftrightarrow$ D27: $+22.5 \leftrightarrow -22.5$

- D7 ↔ D26: $+22.5 \leftrightarrow -22.5$
- D8 ↔ D25: $+22.5 \leftrightarrow -22.5$
- D9 ↔ D24: $+22.5 \leftrightarrow -22.5$
- D10 ↔ D23: $+22.5 \leftrightarrow -22.5$
- D11 ↔ D22: $+22.5 \leftrightarrow -22.5$
- D12 ↔ D21: $+22.5 \leftrightarrow -22.5$
- D13 ↔ D20: $+22.5 \leftrightarrow -22.5$
- D14 ↔ D19: $+22.5 \leftrightarrow -22.5$
- D15 ↔ D18: $+22.5 \leftrightarrow -22.5$
- D16 ↔ D17: $+22.5 \leftrightarrow -22.5$

The mechanical rule never changes: each dual count doubles the domains and creates exactly as many new mirror pairs as there were domains before the split. Opposite domains always mirror each other across the inverse center unit 0. This is the direct cause of the 4D inverse tensor structure across all four faces.

25 The Binary Skeleton — Grouped Slash Notation for Counts and Depths

The unfolding of the ladder is driven by a single hierarchical binary engine. Both the *counts* and the *depths they own* are expressed in the same grouped slash notation. Count 0 begins with the single base slot 0 (the axiom seat, no partition performed). Count 1 introduces the first dual choice 0/1. Each subsequent dual count adds a fresh "00" block (reflecting the simultaneous ownership of one even and one odd depth). The terminal count 5 therefore carries exactly the grouped string

$$\mathbf{0/00/00/00/00}.$$

This string owns the final pair of depths (8 and 9) in the full ten-slot depth register

$$0/0/0/0/0/0/0/0/0/0,$$

where each slot corresponds to one ladder position $d = 0, 1, \ldots, 9$ (slot $d = 0$ is the axiom seat; slots $d = 1$ through $d = 9$ are the nine active depths), and the rightmost bit of any slot is the parity bit forced by the parity rule.

The following table presents the complete binary skeleton in this native grouped notation. Every entry is a direct mechanical consequence of the axiom of 1 and the single parity rule (final bit = 0 for rational, 1 for irrational).

This grouped view makes three facts immediate. First, the domain growth $1 \to 2 \to 4 \to 8 \to 16 \to 32$ and the depth register (one axiom seat plus nine active depths) are two parallel binary ladders locked together by the same slash blocks. Second, the parity rule is literally the final bit of the grouped depth string (0 = rational, 1 = irrational). Third, the entire structure — singular/dual counts, the terminal pair $(16, 16\sqrt{2})$, the machining of 720, the two operational structures, the four

Count	Count bin.	Domains	Depth	Depth binary (10 slots)	Parity	Reading
0	0	1	0	0/0/0/0/0/0/0/0/0/0	even (0)	1
1	0/1	2	1	0/0/0/0/0/0/0/0/0/1	odd (1)	$\sqrt{2}$
2	00	4	2	0/0/0/0/0/0/0/0/1/0	even (0)	2
			3	0/0/0/0/0/0/0/0/1/1	odd (1)	$2\sqrt{2}$
3	0/00	8	4	0/0/0/0/0/0/1/0/0/0	even (0)	4
			5	0/0/0/0/0/0/1/0/0/1	odd (1)	$4\sqrt{2}$
4	0/00/00	16	6	0/0/0/0/0/1/0/0/0/0	even (0)	8
			7	0/0/0/0/0/1/0/0/0/1	odd (1)	$8\sqrt{2}$
5	**0/00/00/00/00**	32	8	0/0/0/0/1/0/0/0/0/0	even (0)	16
			9	0/0/0/0/1/0/0/0/0/1	odd (1)	$16\sqrt{2}$

Table 20: The binary skeleton of the chain expressed in grouped slash notation. Counts and depths share the identical hierarchical structure. Count 5 carries exactly 0/00/00/00/00 and selects the final two slots of the ten-slot depth register (the axiom seat at slot $d = 0$ plus the nine active-depth slots). The rightmost bit of each depth string is forced by the parity rule and determines the rational/irrational reading.

parallel faces, and the 4D inverse tensor of Part Seven — is a necessary consequence of unfolding the ten-slot register under the single parity rule. No additional axioms are required.

The same grouped binary skeleton reappears unchanged in the outside mirror chain (Part Six) and in the inverse-unit indexing of the four terminal faces. It is the mechanical heartbeat shared by every part of the engine.